

Clustering

Unsupervised Learning

Hierarchical Clustering

- Hierarchical clustering refers to a clustering process that organizes the data into large groups, which contain smaller groups and so on.
- A hierarchical clustering may be drawn as a tree or dendrogram.
- The finest grouping is at the bottom of the dendrogram, each sample by itself forms a cluster.
- At the top of the dendrogram, where all samples are grouped into one cluster

Agglomerative Clustering

- Start with the points as individual clusters
- It follows bottom-up approach
- At each step, merge the closest pair of clusters until only one cluster (or k clusters) left.

Ex: single-linkage, complete-linkage, Average linking algorithm

Agglomerative Clustering Algorithm

1. Compute the proximity matrix
2. Let each data point be a cluster
- 3. Repeat**
4. Merge the two closest clusters
5. Update the proximity matrix
- 6. Until** only a single cluster remains

Key operation is the computation of the proximity of two clusters

- Different approaches to defining the distance between clusters distinguish the different algorithms

Some commonly used criteria in Agglomerative clustering Algorithms

(The most popular distance measure used is Euclidean distance)

Single Linkage:

Distance between two clusters is the smallest pairwise distance between two observations/nodes, each belonging to different clusters.

Complete Linkage:

Distance between two clusters is the largest pairwise distance between two observations/nodes, each belonging to different clusters.

Mean or average linkage clustering:

Distance between two clusters is the average of all the pairwise distances, each node/observation belonging to different clusters.

Centroid linkage clustering:

Distance between two clusters is the distance between their centroids.

Agglomerative Clustering (Single-Link)

Use the **Euclidean Distance** to measure proximity between points (x_1, and_1) and (x_2, and_2) :

$$d = \sqrt{(x_2 - x_1)^2 + (\text{and}_2 - \text{and}_1)^2}$$

Dataset of 5 points is given:

Points	X	Y
P1	4	4
P2	8	4
P3	15	8
P4	24	4
P5	24	12

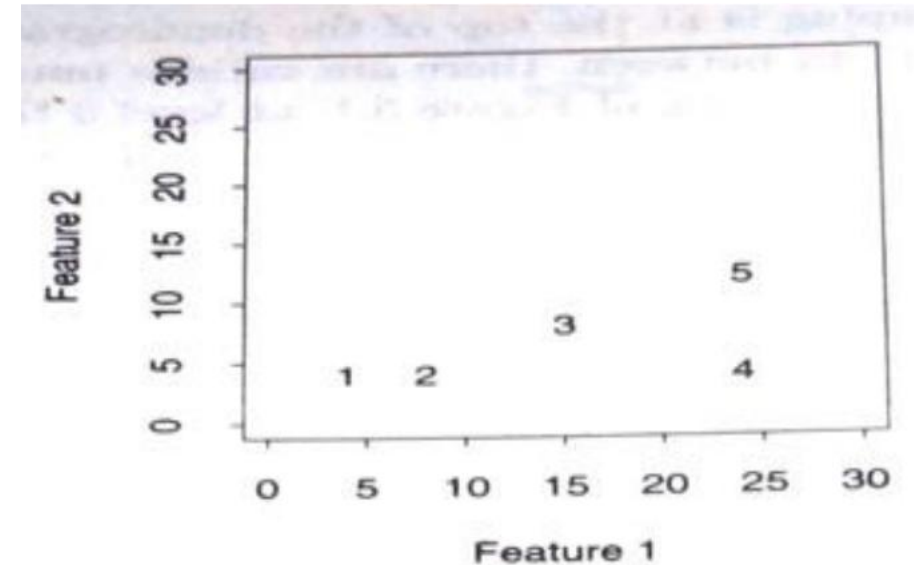


Figure 5.2: Samples for clustering.

Agglomerative Clustering (Single-Link)

Step 1: Calculate Distance Matrix

Points	X	Y
P1	4	4
P2	8	4
P3	15	8
P4	24	4
P5	24	12

$$d(1, 2) = \sqrt{(8 - 4)^2 + (4 - 4)^2} = \sqrt{16 + 0} = 4.0$$

$$d(1, 3) = \sqrt{(15 - 4)^2 + (8 - 4)^2} = \sqrt{121 + 16} = \sqrt{137} \approx 11.7$$

$$d(2, 3) = \sqrt{(15 - 8)^2 + (8 - 4)^2} = \sqrt{49 + 16} = \sqrt{65} \approx 8.06$$

$$d(3, 4) = \sqrt{(24 - 15)^2 + (4 - 8)^2} = \sqrt{81 + 16} = \sqrt{97} \approx 9.85$$

$$d(3, 5) = \sqrt{(24 - 15)^2 + (12 - 8)^2} = \sqrt{81 + 16} = \sqrt{97} \approx 9.85$$

$$d(4, 5) = \sqrt{(24 - 24)^2 + (12 - 4)^2} = \sqrt{0 + 64} = 8.0$$

Agglomerative Clustering (Single-Link)

Step 2: Find Initial Proximity Matrix

Points	X	Y
P1	4	4
P2	8	4
P3	15	8
P4	24	4
P5	24	12

	P1	P2	P3	P4	P5
P1	0	4.0	11.70	20.0	21.54
P2	4.0	0	8.06	16.0	17.89
P3	11.70	8.06	0	9.85	9.85
P4	20.0	16.0	9.85	0	8.0
P5	21.54	17.89	9.85	8.0	0

Agglomerative Clustering (Single-Link)

Smallest distance b/w P1 and P2

Points	X	Y
P1	4	4
P2	8	4
P3	15	8
P4	24	4
P5	24	12

	P1	P2	P3	P4	P5
P1	0	4.0	11.70	20.0	21.54
P2	4.0	0	8.06	16.0	17.89
P3	11.70	8.06	0	9.85	9.85
P4	20.0	16.0	9.85	0	8.0
P5	21.54	17.89	9.85	8.0	0

Agglomerative Clustering

Next smallest distance is b/w P4 & P5. Find distance of (1,2) from Other points using MIN rule:

Points	X	Y
P1	4	4
P2	8	4
P3	15	8
P4	24	4
P5	24	12

	(1,2)	P3	P4	P5
(1,2)	0	8.06	16.0	17.89
P3	8.06	0	9.85	9.85
P4	16.0	9.85	0	8.0
P5	17.89	9.85	8.0	0

Agglomerative Clustering

Updated Matrix after calculating MIN distance of (4,5) from other points:

Points	X	Y
P1	4	4
P2	8	4
P3	15	8
P4	24	4
P5	24	12

	(1,2)	P3	(4,5)
(1,2)	0	8.06	16.0
P3	8.06	0	9.85
(4,5)	16.0	9.85	0

Agglomerative Clustering

Merge (1,2) with P3:

Points	X	Y
P1	4	4
P2	8	4
P3	15	8
P4	24	4
P5	24	12

	(1,2)	P3	(4,5)
(1,2)	0	8.06	16.0
P3	8.06	0	9.85
(4,5)	16.0	9.85	0

Agglomerative Clustering

Merge (1,2) with P3. Updated Matrix is:

Points	X	Y
P1	4	4
P2	8	4
P3	15	8
P4	24	4
P5	24	12

	(1,2,3)	(4,5)
(1,2,3)	0	9.85
(4,5)	9.85	0

Agglomerative Clustering

Merge (1,2) with P3. Updated Matrix is:

Points	X	Y
P1	4	4
P2	8	4
P3	15	8
P4	24	4
P5	24	12

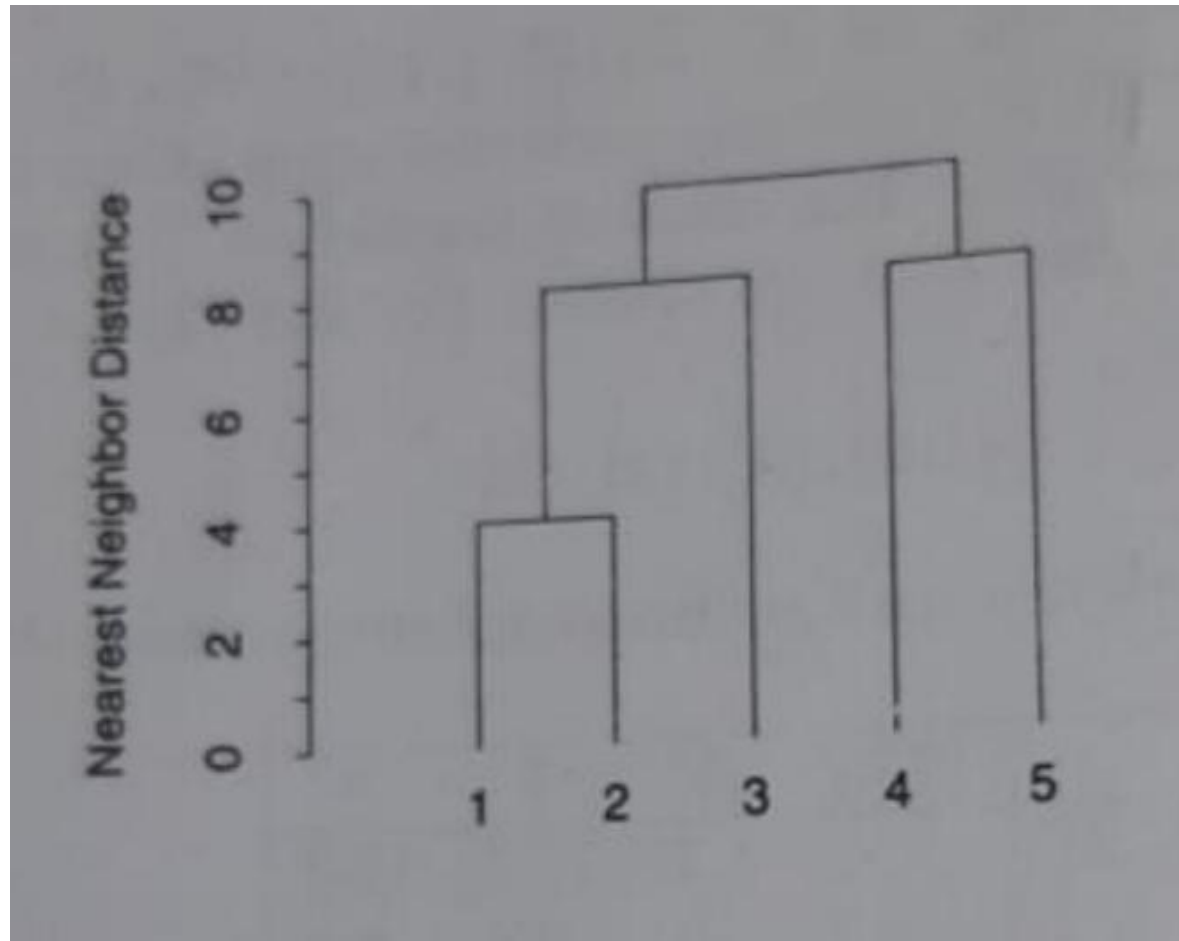
	(1,2,3)	(4,5)
(1,2,3)	0	9.85
(4,5)	9.85	0

At final step, merge (1,2,3) with (4,5)

Agglomerative Clustering

At final step, merge (1,2,3) with (4,5) at distance **9.85**

Points	X	Y
P1	4	4
P2	8	4
P3	15	8
P4	24	4
P5	24	12



Agglomerative Clustering (Complete Linkage)

Merge 1: Use Minimum distance to make First Merge.

Points	X	Y
P1	4	4
P2	8	4
P3	15	8
P4	24	4
P5	24	12

	P1	P2	P3	P4	P5
P1	0	4.0	11.70	20.0	21.54
P2	4.0	0	8.06	16.0	17.89
P3	11.70	8.06	0	9.85	9.85
P4	20.0	16.0	9.85	0	8.0
P5	21.54	17.89	9.85	8.0	0

Agglomerative Clustering (Complete)



Update the distances from **(1,2)** to all other points using the **MAX** rule:

$$d((1,2),3) = \max(d(1,3), d(2,3)) = \max(11.7, 8.06) = \mathbf{11.7}$$

$$d((1,2),4) = \max(d(1,4), d(2,4)) = \max(20, 16) = \mathbf{20.0}$$

$$d((1,2),5) = \max(d(1,5), d(2,5)) = \max(21.54, 17.89) = \mathbf{21.54}$$

Next: Merge P4 & P5?

	(1,2)	P3	P4	P5
(1,2)	0	11.7	20.0	21.54
P3	11.7	0	9.85	9.85
P4	20.0	9.85	0	8.0
P5	21.54	9.85	8.0	0

Agglomerative Clustering (Complete)

Update the distances from **(4,5)** to all other points using the **MAX** rule:

$$d((4, 5), 3) = \max(d(4, 3), d(5, 3)) = \max(9.85, 9.85) = \mathbf{9.85}$$

$$d((4, 5), (1, 2)) = \max(d(4, (1, 2)), d(5, (1, 2))) = \max(20.0, 21.54) = \mathbf{21.54}$$

Next: Merge P3 & (4,5)

	(1,2)	P3	(4,5)
(1,2)	0	11.7	21.54
P3	11.7	0	9.85
(4,5)	21.54	9.85	0

Agglomerative Clustering (Complete)

Update the distances from **(3,4,5)** to all other points using the **MAX** rule:

$$d((1, 2), (3, 4, 5)) = \max(d((1, 2), 3), d((1, 2), 4), d((1, 2), 5))$$

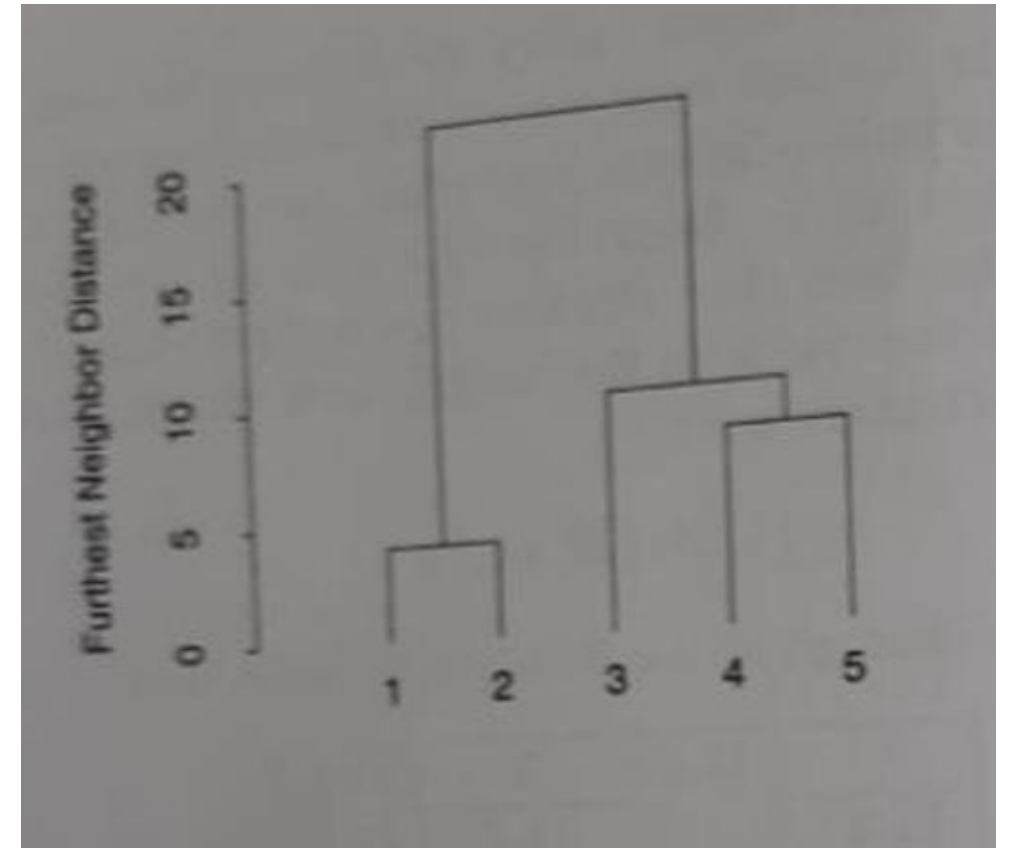
$$d((1, 2), (3, 4, 5)) = \max(11.7, 20.0, 21.54) = \mathbf{21.54}$$

	(1,2)	(3,4,5)
(1,2)	0	21.54
(3,4,5)	21.54	0

Agglomerative Clustering (Complete)

Final Merge, (1,2) merges with (3,4,5)
at distance **21.54**

	(1,2)	(3,4,5)
(1,2)	0	21.54
(3,4,5)	21.54	0



Agglomerative Clustering (Average)

Merge 1: The smallest distance is **4.0**. We merge **P1** and **P2** into cluster **(1,2)**.

Points	X	Y
P1	4	4
P2	8	4
P3	15	8
P4	24	4
P5	24	12

	P1	P2	P3	P4	P5
P1	0	4.0	11.70	20.0	21.54
P2	4.0	0	8.06	16.0	17.89
P3	11.70	8.06	0	9.85	9.85
P4	20.0	16.0	9.85	0	8.0
P5	21.54	17.89	9.85	8.0	0

Agglomerative Clustering (Average)

Next: Update the distances from cluster **(1,2)** to the other points by AVG rule.

$$\text{Dist}((1,2), P3): (11.70 + 8.06)/2 = \mathbf{9.88}$$

$$\text{Dist}((1,2), P4): (20.0 + 16.0)/2 = \mathbf{18.0}$$

$$\text{Dist}((1,2), P5): (21.54 + 17.89)/2 = \mathbf{19.72}$$

	(1,2)	P3	P4	P5
(1,2)	0	9.88	18.0	19.72
P3	9.88	0	9.85	9.85
P4	18.0	9.85	0	8.0
P5	19.72	9.85	8.0	0

Agglomerative Clustering (Average)

Next: The smallest distance is **8.0**. We merge **P4** and **P5** into cluster **(4,5)**.

Dist((4,5), P3): $(9.85 + 9.85)/2 = \mathbf{9.85}$

Dist((4,5), (1,2)): This involves 4 pairwise distances: $(d(1, 4) + d(1, 5) + d(2, 4) + d(2, 5))/4$.

- $(20.0 + 21.54 + 16.0 + 17.89)/4 = 75.43/4 = \mathbf{18.86}$

	(1,2)	P3	(4,5)
(1,2)	0	9.88	18.86
P3	9.88	0	9.85
(4,5)	18.86	9.85	0

Agglomerative Clustering (Average)

Next: The smallest distance is **9.85**. We merge **P3** with cluster **(4,5)** to form cluster **(3,4,5)**.

$$\text{Dist}((1,2), (3,4,5)): (d(1, 3) + d(1, 4) + d(1, 5) + d(2, 3) + d(2, 4) + d(2, 5))/6$$
$$(11.70 + 20.0 + 21.54 + 8.06 + 16.0 + 17.89)/6 = 95.19/6 = \mathbf{15.87}$$

	(1,2)	(3,4,5)
(1,2)	0	15.87
(3,4,5)	15.87	0

Agglomerative Clustering (Average)

Next: The two clusters merge at **Height 15.87**.

Find Dandograms illustration of Average Clustering?